

T1 Rank as a selection rule against the published alternatives - and it is UNDERPOWERED

Does the metric pick the arm that carries more molecular information? Ground truth: held-out top-CCA on the 40 targets neither arm trained on, confound-residualised block.

n = 6. Exact two-sided binomial: 6/6 -> p = 0.031; 5/6 -> 0.219; 4/6 -> 0.688.
THIS DESIGN CAN DETECT A PERFECT RULE AND NOTHING WEAKER. No comparison between two rows of this table is evidence in either direction.

metric	D2 s42	D2 s43	D2 s44	D1 s42	D1 s43	D1 s44	D2	D1	ALL	exact two-sided p
effective_rank_raw	✓	✗	unres.	✓	✓	✓	2/3	3/3	5/6	0.219
effective_rank_residualised	✓	✗	unres.	✓	✓	✓	2/3	3/3	5/6	0.219
rankme_raw (as published)	✓	✓	unres.	✗	✓	✗	3/3	1/3	4/6	0.688
rankme_residualised	✓	✗	unres.	✓	✓	✓	2/3	3/3	5/6	0.219
participation_ratio_raw	✓	✓	unres.	✗	✓	✓	2/3	2/3	4/6	0.688
participation_ratio_residualised	✓	✓	unres.	✗	✓	✓	2/3	2/3	4/6	0.688
stable_rank_raw	✓	✓	unres.	✗	✓	✗	2/3	1/3	3/6	1.000
stable_rank_residualised	✗	✓	unres.	✗	✓	✓	1/3	2/3	3/6	1.000
alpha_ReQ_raw alpha-1 <i>authors' released fastssl estimator, not the paper text</i>	✓	✗	unres.	✓	✓	✗	2/3	2/3	4/6	0.688
alpha_ReQ_resid alpha-1 <i> alpha-1 is OUR operationalisation of their "Goldilocks zone"</i>	✓	✗	unres.	✓	✓	✗	2/3	2/3	4/6	0.688
LiDAR_raw <i>ADAPTED: q = 2, the two modalities as views</i>	✗	✗	unres.	✓	✓	✓	0/3	3/3	3/6	1.000
LiDAR_residualised <i>outside the authors' tested q range; ordering invariant over delta 1e-8 to 1e0</i>	✗	✓	unres.	✓	✓	✓	1/3	3/3	4/6	0.688

D2 s44 IS MARKED UNRESOLVABLE IN EVERY ROW. F4(a) puts that pair's between-arm gap at 1.4 sampling standard deviations under 40 draws of 80% of patients, so no row of this table can score it. Effective rank's honest D2 record therefore reads: 1 clear hit, 1 clear miss, 1 pair it cannot resolve.
The counts in the D2, D1 and ALL columns are the run's own, which score that cell; they are printed as the run produced them and are NOT corrected here.

LiDAR is ADAPTED - q = 2 with the two modalities as views, licensed by the paper's own footnote 4 but outside the authors' tested q range - and its ordering is invariant across delta from 1e-8 to 1e0. alpha-ReQ follows the authors' released fastssl estimator rather than the paper text, and |alpha - 1| is OUR operationalisation of their "Goldilocks zone", not theirs: all twelve artifacts sit at alpha between 2.6 and 4.8, far outside it. RankMe as published is uncentred on the RAW block, per its own definition; no value in this table is compared against a PUBLISHED RankMe number.